

A university wants to analyze the **career pathways of students** after graduation. The available dataset records the flow of students from their **academic departments** to **specializations** and finally to their **career outcomes**.

The dataset contains the following hierarchy:

- **Departments:** Computer Science, Information Technology, Electronics, Mechanical, Civil
- **Specializations:** AI & ML, Data Science, Cyber Security, Embedded Systems, Manufacturing, Construction
- **Career Outcomes:** Software Engineer, Data Analyst, Research, Higher Studies

The relationships between these categories are stored as **source node, target node, and flow value**, where the flow value represents the number of students moving from one stage to the next.

Using **R** and the **networkD3** package, answer the following:

1. Create a **Sankey Plot** representing the complete flow of students from departments to specializations and career outcomes.
2. Configure the diagram with:
 - Node width = **35**
 - Font size = **14**
 - Career outcomes displayed on the **right side** of the diagram.
3. Interpret the visualization and identify:
 - The specialization receiving the **highest number of students**.
 - The career outcome with the **largest overall inflow**.
 - Which department contributes the most students to **Software Engineer** positions.
4. Explain why a **Sankey diagram** is more appropriate than a bar chart or pie chart for visualizing this dataset.

5. Suggest **two improvements** that would make the visualization more informative for university administrators.

Expected Output:

A Sankey diagram showing the flow of students from **Departments** → **Specializations** → **Career Outcomes**, along with a brief interpretation of the major pathways and student distribution.

Database:

ID	Node
0	Computer Science
1	Information Technology
2	Electronics
3	Mechanical
4	Civil
5	AI & ML
6	Data Science
7	Cyber Security
8	Embedded Systems
9	Manufacturing
10	Construction
11	Software Engineer
12	Data Analyst
13	Research
14	Higher Studies

Link table:

Source	Target	Value
0	5	80
0	6	70
0	7	50
1	6	60
1	7	50
2	8	90
3	9	100
4	10	80
5	11	50
5	13	30
6	11	40
6	12	60
6	14	30
7	11	35
7	13	25
8	13	40
8	14	50
9	11	70
10	11	50

Source	Target	Value
10	14	30

R Program (networkD3 Sankey Diagram)

Install package (Run once)

```
install.packages("networkD3")
```

```
library(networkD3)
```

```
nodes <- data.frame(
```

```
  name = c(
```

```
    "Computer Science",
```

```
    "Information Technology",
```

```
    "Electronics",
```

```
    "Mechanical",
```

```
    "Civil",
```

```
    "AI & ML",
```

```
    "Data Science",
```

```
    "Cyber Security",
```

```
    "Embedded Systems",
```

```
    "Manufacturing",
```

```
    "Construction",
```

```
    "Software Engineer",
```

```
    "Data Analyst",
```

```
    "Research",
```

```
    "Higher Studies"
```

```

)
)
links <- data.frame(
  source = c(0,0,0,1,1,2,3,4,5,5,6,6,6,7,7,8,8,9,10,10),
  target = c(5,6,7,6,7,8,9,10,11,13,11,12,14,11,13,13,14,11,11,14),
  value = c(80,70,50,60,50,90,100,80,50,30,40,60,30,35,25,40,50,70,50,30)
)
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  nodeWidth = 35,
  fontSize = 14,
  sinksRight = TRUE
)

```

Interpretation

1. Specialization receiving the highest number of students

Calculate total inflow to each specialization.

Specialization	Students
AI & ML	80

Specialization	Students
Data Science	$70 + 60 = \mathbf{130}$
Cyber Security	$50 + 50 = \mathbf{100}$
Embedded Systems	90
Manufacturing	100
Construction	80

Answer: Data Science receives the highest number of students (**130 students**).

2. Career outcome with the largest overall inflow

Career Outcome	Total Students
Software Engineer	$50 + 40 + 35 + 70 + 50 = \mathbf{245}$
Data Analyst	60
Research	$30 + 25 + 40 = \mathbf{95}$
Higher Studies	$30 + 50 + 30 = \mathbf{110}$

Answer: Software Engineer has the largest inflow (**245 students**).

3. Department contributing the most Software Engineers

Tracing the paths:

- Computer Science
 - AI & ML $\rightarrow$ Software Engineer = 50
 - Data Science $\rightarrow$ Software Engineer = 40
 - Cyber Security $\rightarrow$ Software Engineer = 35
 - **Total $\approx$ 125**

- Information Technology
 - Data Science → Software Engineer (part)
 - Cyber Security → Software Engineer (part)
- Mechanical
 - Manufacturing → Software Engineer = 70
- Civil
 - Construction → Software Engineer = 50
- Electronics
 - Embedded Systems does not connect to Software Engineer.

Answer: Computer Science contributes the highest number of students to **Software Engineer** careers.

Why is a Sankey Diagram better?

A Sankey diagram is more suitable because:

- It clearly shows the **flow** of students from departments to specializations and then to career outcomes.
 - The width of each link represents the **number of students**, making major pathways easy to identify.
 - It visualizes **multiple stages** simultaneously.
 - A **bar chart** only compares totals and cannot show relationships between stages.
 - A **pie chart** only shows proportions and cannot represent flows or transitions.
-

Two Improvements

1. Use **different colors** for each department and specialization to make pathways easier to distinguish.

2. **Add interactive tooltips** showing the exact number of students for each flow when the user hovers over a link or node.

Output:

